

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf _____

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? _____Serial0/0/0_____

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? _____129_____

Does R2 show up as an OSPF neighbor on R1? ____Yes____

Does R2 show up as an OSPF neighbor on R3? ____No____

What does this information tell you?

R2's S0/0/1 interface to R3 is passive, so no OSPF adjacency forms with R3 and no Hellos are exchanged. However, the 192.168.2.0/24 network is still reachable on R3 via the alternate path through R1. This shows that passive interfaces block adjacency on that link but do not prevent route propagation through other neighbors.

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

R2(config)# router ospf 1 _____

R2(config-router)# no passive-interface s0/0/1 _____

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? _____Serial0/0/1_____

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?
cost of R3's outgoing interface S0/0/1 (64) + cost of R2's G0/0 LAN interface (1) = 65 _____

Is R2 listed as an OSPF neighbor to R3? ____Yes____

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

To accurately differentiate high-speed links. The default 100 Mbps reference bandwidth assigns a cost of 1 to all links ≥ 100 Mbps. Increasing it gives lower/faster-link costs, producing more accurate shortest-path calculations.

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

192.168.3.0/24 cost = 782 = cost of R1 S0/0/1 (781) + cost of R3 G0/0 (1).

192.168.23.0/30 cost = 845 = cost of R1 S0/0/1 (781) + cost of R3 S0/0/1 (64).

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

$2 \times (100 / 0.128) = 2 \times 781 = 1562$.

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

The manual ip ospf cost 1565 on R1's S0/0/1 makes the direct path cost 1565 (R1) + 1 (R3 LAN) = 1566. The alternate path via R2 is lower: 781 (R1 S0/0/0) + 781 (R2 S0/0/1) + 1 (R3 LAN) = 1563. OSPF always prefers the lower-cost path.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

To ensure the router ID is unique and stable across the OSPF domain. Without explicit control, it may change if interface IPs change, breaking adjacencies or causing duplicate-ID issues.

2. Why is the DR/BDR election process not a concern in this lab?

All inter-router links are point-to-point serial. LANs (G0/0) each have only one router, so no multi-access network exists requiring election.

3. Why would you want to set an OSPF interface to passive?

To stop Hello packets and adjacency formation on interfaces connected only to end hosts. This reduces unnecessary traffic/CPU load and prevents routing updates from reaching non-OSPF devices.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

Multi-area design with area 0 as the backbone. It limits LSA flooding, enables route summarization at ABRs, and reduces routing table size in large networks.

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

network 10.0.0.0 0.255.255.255 area 0

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

The EIGRP route. Administrative distance is checked first: EIGRP (90) < OSPF (110) < RIP (120). The lowest AD wins regardless of metric.